

Aakash Siricilla

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years designing, operating, and scaling distributed backend and AI/ML systems in production. Builds high-throughput Java/Spring Boot and Python/FastAPI services, event-driven Kafka/AWS pipelines, and LLM/RAG and anomaly-detection workloads served behind SLOs. Owns systems end to end: capacity planning, Kubernetes deployment, CI/CD, observability, on-call, and incident root-cause analysis. M.S. in Computer Science. Creator of Astra, an autonomous LLM computer-use agent. Targeting AI/ML Systems, SDE, and Systems Engineering roles.

TECHNICAL SKILLS

Languages	Java, Python, TypeScript/JavaScript, SQL, Bash/Shell, C/C++
AI/ML Systems	PyTorch, scikit-learn, XGBoost, LLMs, RAG, Embeddings & Vector Search (Qdrant, FAISS), Agentic Workflows (Tool Use, Computer Use), Model Serving, MLflow, Evaluation Harnesses, Drift Monitoring
Backend Systems	Spring Boot, FastAPI, REST/gRPC/WebSocket APIs, Microservices, Kafka, Celery, Event-Driven Processing, Caching, Idempotency, Fault Tolerance, Capacity Planning
Data & Storage	PostgreSQL, MySQL, Redis, TimescaleDB, SQLite, ETL/Streaming Pipelines, Query Optimization, Schema Design
Cloud & Ops	AWS (EC2, EKS, S3, RDS, Lambda, SQS, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Linux, Prometheus, Grafana, SLOs/SLIs, On-Call, Incident Response, Runbooks
Frontend/Testing	React, Next.js, Tailwind CSS, JUnit, Mockito, pytest, Load Testing (k6), Git

PROFESSIONAL EXPERIENCE

AT&T <i>Software Engineer, Distributed Systems</i>	Sep 2024 – Present <i>Remote, USA</i>
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- Designed and implemented an event-driven service-assurance platform (Java, Spring Boot, Kafka, PostgreSQL) ingesting ~40M network and customer-operations events/day from 12+ upstream systems, replacing batch reconciliation and cutting manual effort 40%.
- Built high-throughput REST/gRPC APIs sustaining ~2.5k RPS with connection pooling, Redis caching, and idempotent writes; rewrote hot-path PostgreSQL queries with composite indexes and query-plan tuning, reducing p95 latency by 35% (480ms → 310ms).
- Deployed and maintained services across dev/stage/prod on AWS EKS with Docker, Terraform, and GitHub Actions CI/CD (canary releases, automated rollbacks, config management), taking release-related incidents from ~3/month to near zero.
- Built an ML anomaly-detection service (Python, scikit-learn, XGBoost, FastAPI) scoring streaming operational metrics against seasonal baselines, with MLflow-tracked retraining and drift monitoring; cut false pages 30% and time-to-detect from hours to <10 min.
- Instrumented services with Prometheus/Grafana and CloudWatch, defined SLOs/SLIs (99.9% availability, p95 latency) with burn-rate alerts, and wrote runbooks; on-call rotation member resolving tickets within SLOs and leading P1/P2 root-cause analysis.
- Automated operations with Python and Bash: health checks, log parsing, data validation, backfills, and scheduled maintenance, eliminating ~15 hours/week of manual runbook steps and reducing on-call toil.
- Partnered with product, network-operations, and security teams to define distributed-systems requirements, capacity plans, and test procedures; delivered role-based access control and transactional audit logging, reducing investigation time 45%.

Flipkart <i>Software Engineer</i>	Apr 2021 – Sep 2023 <i>Hyderabad, India</i>
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- Developed customer-facing CRM and workflow-automation services in Java, Spring Boot, React, and PostgreSQL used by 1,500+ internal agents for lead management and client onboarding, reducing manual coordination effort by 35%.
- Designed REST APIs, service-layer validation, and data-access logic for transaction-heavy flows handling ~5M records/day; added Redis caching and removed N+1 queries, improving retrieval performance 25% and holding p99 under target at festive-sale peaks.
- Built asynchronous background-job processing (Kafka consumers and scheduled workers) for notifications, exports, and batch updates, decoupling long-running work from request threads and raising job throughput 3x with retry and dead-letter handling.
- Built real-time monitoring dashboards and alerting for API health, queue lag, and error rates; resolved production tickets and bugs within service-level objectives and shipped fixes through CI with JUnit/Mockito coverage above 80%.
- Created a reusable React component library integrated with Spring Boot services, reducing feature delivery time 30% and standardizing UI across four teams.

EDUCATION

New Jersey Institute of Technology , Master of Science in Computer Science Coursework: Distributed Systems, Deep Learning, Machine Learning & MLOps	Sep 2023 – May 2025
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Kommuri Pratap Reddy Institute of Technology , B.Tech. in Computer Science & Engineering	Aug 2018 – May 2022
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PROJECTS

- Astra (ApplyPilotAI) – Autonomous LLM Computer-Use Agent** | Python, FastAPI, Chrome MV3, Electron, SQLite | GitHub
- Architected an agent that completes real digital tasks (job applications, forms, app work) via a screenshot → model action → execute → read-back verification loop, with swappable browser (DOM-verified Chrome extension) and desktop (UIA/SendInput) execution bodies.
 - Built the safety and reliability layer: approval gates before any submit/pay, a credential vault the model never sees, sensitive-field masking in screenshots, URL allow-listing, per-step receipts, and a FastAPI/SQLite backend covered by ~1,200 tests.
- DocuMindAI – Document Intelligence (RAG) Platform** | FastAPI, React, Qdrant, Redis, Celery, MLflow, Docker | GitHub
- Built a RAG platform with async ingestion, chunking/embedding pipelines, hybrid (dense + keyword) retrieval over Qdrant, and LLM chat; indexing offloaded to Celery/Redis workers, retrieval experiments tracked in MLflow; cut lookup time ~70%.
- SyncStream – Real-Time Collaborative Streaming** | Java, Spring Boot, Next.js, WebSockets, PostgreSQL | GitHub
- Built a real-time collaboration platform with invite-based sessions, persistent room state, and low-latency WebSocket sync; Dockerized.